```
import pandas as pd
import requests
from datetime import datetime, timezone
import urllib3
urllib3.disable_warnings(urllib3.exceptions.InsecureRequestWarning)
```

```
def fetch_exchange_rates():
  url ="https://open.er-api.com/v6/latest/USD"
  data= requests.get(url,verify=False).json()
  data["conversion_rates"]= data["rates"]
  print(f"Fetched {len(data['conversion_rates'])} currencies")
  return data
raw_data = fetch_exchange_rates()
```

```
Fetched 166 currencies
```

```
def extract_currencies(raw_data):
  records=[{"currency_code": c, "rate": float(r),
            "extracted_at": datetime.now(timezone.utc).isoformat()}
           for c, r in raw_data["conversion_rates"].items()]

  print(f" Extraded {len(records)} curriencies")
  return records
records=extract_currencies(raw_data)
```

```
 Extraded 166 curriencies
```

```
def clean_currencies(records):
  dupli = ["GGP","IMP","JEP","FKP","KID","TVD"]
  cleaned= [r for r in records
            if r["rate"] > 0
            and r["currency_code"] not in dupli]
  print(f"Cleaned: {len(cleaned)} currencies")
  return cleaned

cleaned_records = clean_currencies(records)
```

```
Cleaned: 160 currencies
```

```
Start coding or generate with AI.
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